THE CARBON CYCLE

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Humans are sometimes described as **carbon-based life forms**. Carbon dioxide (CO_2) is an important way of transferring carbon compounds between living things and their environment. Carbon is one element that is constantly being removed from compounds that are being broken down to be used to make new compounds. This process can be described as the **carbon cycle**. Part of the carbon cycle is shown in figure 1.

1 List the processes that add CO_2 to the atmosphere.

2 List the processes that remove CO_2 from the atmosphere.

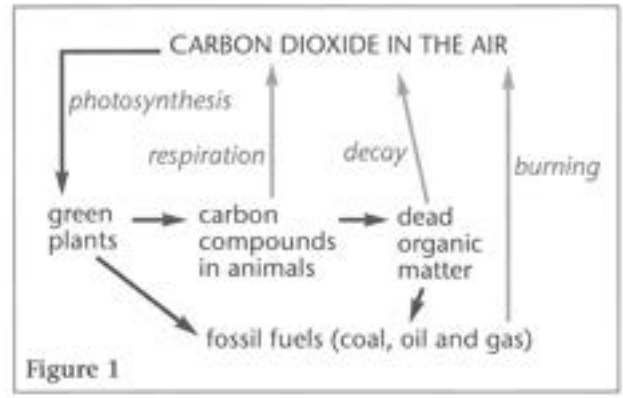


Figure 1

Photosynthesis takes place in the leaves of a plant. This process requires heat and light energy from sunlight. Chlorophyll must also be present.

3 Define *photosynthesis*.

4 What role does chlorophyll play in photosynthesis?

5 What raw materials are needed for photosynthesis?

Inside a plant the simple glucose molecules ($C_6H_{12}O_6$) can be joined together to form starch, cellulose and other complex carbon compounds. Animals then obtain carbon compounds (and energy) by eating plants. Humans obtain carbon compounds by eating both animals and plants. All food on Earth ultimately derives its energy from the Sun. Only about 1% of the solar energy that hits a green plant is converted to stored (chemical) energy. If these plants are not eaten directly by humans but used as fodder for animals used for meat, then more energy is 'lost' to us because some energy is used up by the animal. About one-tenth of the energy in the plants eaten by cattle is converted into stored energy (i.e. meat).

6 What percentage of solar energy is turned into stored energy in an animal used for meat?

If plants are allowed to rot and then compressed for a long period of time (millions of years), the carbon compounds eventually change to coal. Crude oil (petroleum) and natural gas (methane) are formed in a similar way from other organic compounds.

7 Approximately how long does coal take to form?

Respiration is the process by which plants and animals give out carbon dioxide. It is the reaction that living things use to get the energy they need. Respiration releases as heat, the energy (originally from sunlight) that is stored as chemical energy in glucose. Respiration takes place in cells. The reaction can be written as: glucose + oxygen → carbon dioxide + water + energy.

8 Write a balanced equation for respiration.

9 Photosynthesis is an exothermic reaction. What is meant by *exothermic*?

Bacteria and other decomposers feed on dead plants and animals. Carbon dioxide is produced at the same time. Burning fuels also produces carbon dioxide.

10 Name three common fuels.

Carbon dioxide contributes to the **Greenhouse effect**. This effect keeps in heat and helps keep the Earth within a steady temperature range. Many people are concerned that we are burning too much fossil fuel and that this will increase the amount of carbon dioxide in the atmosphere.

11 What could too much carbon dioxide in the atmosphere do?

12 Name two substances that plants make using glucose.

13 What is the role of decomposers in the carbon cycle?

14 Mark each statement as true or false.

a Respiration describes the synthesis of complex organic compounds.

b Photosynthesis is a heat-absorbing process.

c Glucose is the base 'ingredient' for starch formation.

d All living things contain carbon.

Research 15 Oceans have not been mentioned on this page. Outline how oceans (and the organisms living in them) contribute to the level of carbon dioxide found in the air.

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